SIH 2026 · ISRO · PS26171 · TEAM WORKFLOW

2 Perception & Local Inference

Owns: *DOM/A11y capture, screen capture, ONNX Runtime Web, WebGPU→WASM→CPU fallback, picking/quantizing the vision model.*

Role 2 of 6 · PERCEPTION

HW Hardware access

Heavy. You hold standing priority on Laptop A for the whole project.

Do initial model comparisons on Colab/Kaggle before ever touching the laptop — reserve physical laptop time for what must run locally: real in-browser WebGPU benchmarking.

P0 Phase 0 — now → 20 Sep 2026 (lock the idea)

- Get access to Laptop A (or a cloud GPU notebook) and benchmark 2–3 candidate lightweight vision models against the official “visual context accuracy” metric.
- Spike ONNX Runtime Web + WebGPU in a bare test page to confirm feasibility before it's claimed publicly in the pitch.

P1 Phase 1 — build the MVP (20 Sep → shortlist)

- DOM/accessibility capture — normalize page state (element roles, labels, ARIA, form types) into one structured representation.
- Screen/visual capture wired to the local model, with runtime capability detection selecting WebGPU → WASM → CPU.
- Log the selected inference mode and measured latency on **every** run — Integration's benchmark harness needs this from day one, not bolted on later.
- Hand off your state-hash format to PVM & Verification early — they key their entire memory store off it.

P2 Phase 2 — rehearse & harden (shortlist → Dec finale)

- Run the full hardware matrix — both powerful laptops plus at least one low-resource/integrated-graphics machine — and record real numbers.
- Never publish an assumed latency or accuracy figure; only measured ones make it into the demo.

DEP Dependencies

YOU DEPEND ON	Extension (content-script scaffold to run inside), Integration (the mock site to perceive).
DEPENDS ON YOU	Privacy Guard (visual detection needs your capture pipeline), Server AI (needs your sanitized visual context), PVM (needs your state representation to hash).

Full team plan, hardware-booking rules, and the master phase timeline: see the companion *PS26171 Team Workflow & Hardware Plan* document.